

BIRDS OF WAR

Mired in the fog of warfare, the evolution of drones has been quite noteworthy

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When we decided to make drones the focus of the November issue of Digit, a debate soon broke out within our offices - about what a drone exactly is. Throughout human history, there have been a lot of different advancements and directions taken to develop unmanned vehicles for land, air and water. However, today, the term drone is popularly synonymous for unmanned aerial vehicles, with unmanned aquatic vehicles slowly being accepted, with the drone barges of SpaceX getting media coverage. Going after the broader definition of every uncrewed vehicle being a drone would bring autonomous cars, a multitude of spacecraft and lethal autonomous weapons systems as well. Autonomous vehicles, started off with the humble balloon.

OF BALLOONS AND BOATS

The earliest known instance of unmanned aerial vehicles in warfare dates back to the mid-19th century, when Austria attacked Venice using about 200 incendiary balloons that were floated with approximately 20-30 pounds of explosives each. While there are records of some of these balloons being successful, quite a few of them missed their mark due to wind

direction changes, even flowing back into Austrian territory.

The next significant step towards UAVs came after the successful invention of the aeroplane and its subsequent popularisation in warfare. By the end of the first World War, everything from radio technology to modified aeroplanes had been tried out in a bid to build an unmanned aerial vehicle. Successful gyroscope based demonstrations led to the development of the Kettering Bug, the first aerial torpedo in 1918. However, subsequent developments happened after the war.

During the interwar period, multiple successful demonstrations of unmanned aircraft led to a strong focus on developing radio technology for accurate control in the 1930s. In 1935, it was the British RC target ship, the 'DH.82B Queen Bee', from which the term 'drone' is supposed to have become ubiquitous.

Across the pond, Reginald Denny was working on his own vision for drones. He established Reginald Denny Industries in 1934, and evolved his shop to the 'Radioplane Company'. By 1940, his company had won a contract to manufacture RC-4 for the army during World War II.

However, it was Edward M Sorensen who can be credited with the invention of the modern UAV, since his

design could communicate altitude, climbing, banking, direction, rpm etc from a ground station. Without this, early RC UAVs would be blind.

EYES IN THE SKY

Back in 1941, when military forces around the world were trying to figure out how to make better UAVs, the Naval Aircraft Factory in the US had designed a drone, dubbed 'Project Fox', that had a camera on it. This camera was connected to a screen on another aircraft. While its test runs were mostly successful and led to the further development of the idea, disagreements between the type and scale of testing to be done caused it to be further delayed beyond the end of World War II.

Immediately after World War II was over, there was no particularly significant development for UAVs except for a general increase in speed and accuracy designed to keep up with similar progress for aircraft - for which they were often used in target practice. During the cold war era, the use of spy drones shot up, with both the US and USSR indulging in spying upon each other and other countries. Many of these projects were shrouded in secrecy, however, some like the Ryan Firebee, the Lockheed D21, The Lightning Bug series were